



Alberta Public Health Disease Management Guidelines

Coronavirus – MERS/SARS



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Case Definition – Middle East Respiratory Syndrome (MERS)

Confirmed Case

A person with laboratory confirmation of infection^(A) at Canada's National Microbiology Laboratory (NML)

Probable Case

- A person epidemiologically-linked through close contact^(B) to a laboratory-confirmed case and meeting illness criteria^(C) but for whom laboratory diagnosis of MERS-CoV is not available^(D) or negative (if specimen quality or timing is suspect)

OR

- A person meeting who meets exposure^(E) and illness criteria^(C) and for whom laboratory screening test for MERS-CoV was positive but not confirmed by the NML^(F)

Person Under Investigation

A person who meets both exposure^(E) and illness^(D) criteria

Note: Limited data suggests that MERS-CoV can present as a co-infection with other viral pathogens. The identification of one causative agent should not exclude MERS-CoV where the index of suspicion may be high.

^(A) Laboratory confirmation of infection: detection of the virus using MERS-specific nucleic acid amplification tests (NAATs) and/or sequencing and through virus isolation in tissue culture.

^(B) A **close contact** is defined as a person who provided care for the patient, including health care workers (except those wearing appropriate PPE), family members or other caregivers, or who had other similarly close physical contact **OR** who stayed at the same place (e.g., lived with or otherwise had close prolonged contact within two metres) as a probable or confirmed case while the case was ill.

^(C) **Illness criteria:** Illness onset is defined by the earliest start of respiratory symptoms associated with the current episode. Focus is on the detection of severe acute respiratory illness (SARI) defined primarily by respiratory symptoms, i.e., fever (over 38 degrees Celsius) AND new onset of (or exacerbation of chronic) cough or breathing difficulty as well as clinical, radiological or histo-pathological evidence of pulmonary parenchymal disease (e.g., pneumonia, pneumonitis, or Acute Respiratory Distress Syndrome [ARDS]), typically associated with the need for hospitalization, intensive care unit monitoring and/or other severity marker (such as death).

Many infectious diseases present with a spectrum of illness, including mild or asymptomatic infection. Atypical MERS-CoV presentation with absent respiratory symptoms has been documented in the presence of comorbidity, notably immuno-suppression. **Therefore, clinician and public health judgment should be used in assessing patients with milder or atypical presentations where - based on contact, comorbidity or cluster history - the index of suspicion may be raised.** Additional information can be found in the Public Health Agency of Canada (PHAC) [Interim Guidance For Containment When Imported Cases With Limited Human-To-Human Transmission Are Suspected/Confirmed In Canada](#).

^(D) Laboratory confirmation not available due to: (1) no possibility of acquiring samples for laboratory testing for MERS-CoV either because the patient or samples are not available; or (2) laboratory diagnosis negative (i.e., negative MERS-CoV result but specimen quality or timing is suspect).

^(E) **Exposure criteria:** Links within 14 days⁽ⁱ⁾ prior to illness^(C) onset to affected areas⁽ⁱⁱ⁾ (i.e., residence, travel history) OR close contact^(B) with a confirmed or probable case of MERS-CoV or a traveller or resident with any acute respiratory illness returning from an affected area⁽ⁱⁱ⁾. Factors that raise the index of suspicion should also be considered⁽ⁱⁱⁱ⁾.

⁽ⁱ⁾ Incubation period for MERS-CoV is still largely unknown but has been reported as prolonged in one documented instance of person-to-person nosocomial transmission (nine to 14 days). SARS-CoV also demonstrated a prolonged incubation period (median four to five days; range two to 10 days) compared to other human coronavirus infections (average two days; typical range 12 hours to five days). Allowing for inherent variability and recall error and to establish consistency with other emerging respiratory virus monitoring, exposure history based on the prior 14 days is a reasonable and safe approximation.

⁽ⁱⁱ⁾ As affected areas are subject to change, consult the PHAC [Summary of Assessment of Public Health Risk to Canada Associated with MERS-CoV](#) for the most up-to-date information.

⁽ⁱⁱⁱ⁾ Factors that raise the index of suspicion include having a history of being in a healthcare facility (as a patient, worker or visitor) OR having contact with camel or camel products (e.g. raw milk or meat, secretions or excretions, including urine), in an affected area within 14 days of illness onset.

^(F) A positive screening test for MERS-CoV should meet one of the following conditions: (1) a positive PCR result for at least two different specific targets on the MERS-CoV genome; or (2) one positive PCR result for a specific target on the MERS-CoV genome and MERS-CoV sequence confirmation from a separate viral genomic target. Laboratory findings may take up to seven days from specimen submission. See additional notes under Person Under Investigation.

Case Definition – Severe Acute Respiratory Syndrome (SARS)

Confirmed Case

- A person with laboratory evidence infection at NML^(G) **AND** early presentation of clinically compatible signs and symptoms of SARS with or without radiographic evidence consistent with SARS

OR

- A deceased person with **ALL** of the following:
 - A history of early presentation of clinically compatible signs and symptoms of SARS (i.e., fever AND cough OR difficulty breathing resulting in death)
 - Autopsy findings consistent with SARS – evidence of pneumonia or Acute Respiratory Infection (ARI) without an alternate identifiable cause
 - Laboratory evidence of SARS-CoV Infection^(A)

Probable Case

In the absence of laboratory evidence, a person with:

- Early presentation of clinically compatible signs and symptoms of SARS with or without radiographic evidence consistent with SARS

AND

- An epidemiologic link to a person or place linked to SARS, including **AT LEAST ONE** of the following:
 - Close contact* with a confirmed SARS case, within 10 days of onset of symptoms
 - Close contact with a symptomatic person who has laboratory evidence of SARS-CoV infection, within 10 days of onset of symptoms
 - Residence, recent travel or visit to an “area with recent local transmission of SARS” within the 10 days prior to onset of symptoms
 - Close contact with a probable case who has been to an “area with recent local transmission of SARS” within the 10 days prior to onset of symptoms (this includes health care workers who were not wearing personal protective equipment)
 - Laboratory exposure to SARS-CoV where appropriate barriers and personal protective equipment were not in place

OR

A deceased person with **ALL of the following**:

- A history of early presentation of clinically compatible signs and symptoms of SARS
- Autopsy findings consistent with SARS
- An epidemiologic link to a person or place linked to SARS

^(G) Laboratory confirmation should involve **ONE** of the following:

- detection of SARS-CoV RNA in appropriate samples (with the caveat of confirmation by NML or a designated laboratory)
- serologic detection of SARS-CoV in a convalescent sample taken > 28 days after onset of illness
- seroconversion between acute and convalescent blood samples collected at least four weeks apart

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed, probable and person under investigation (MERS only) cases in the prescribed form by the Fastest Means Possible (FMP).

Laboratories

All laboratories shall report all positive laboratory results by FMP (i.e., by direct voice communication) to:

- the MOH (or designate) of the zone, and
- the Chief Medical Officer of Health (CMOH) (or designate).

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- The MOH (or designate) of the zone where the case currently resides shall notify the CMOH (or designate) by FMP (i.e., by direct voice communication) of all confirmed and probable cases.
- The MOH (or designate) of the zone where the case currently resides shall forward the [Emerging Respiratory Pathogens and Severe Acute Respiratory Infection \(SARI\) Case Report Form](#) of all confirmed and probable cases (including deaths) to the CMOH (or designate) within 24 hours of notification and the final report within one week of discharge from hospital or resolution of the case status.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by FMP (i.e., by direct voice communication):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Updated web links
	Case Definition	• Minor updates to MERS case definition to better align with national case definition.